

Known Properties of Vacuum Energy, Dark Matter and JWST Early Galaxy Formation

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Abstract

The James Webb Space Telescope has revealed massive galaxies and supermassive black holes at redshifts $z > 10$, within 400 Myr of the Big Bang—structures that Λ CDM struggles to produce in the available time. We propose a resolution based on the separation of two processes that are routinely conflated: (1) the expansion of space, driven by the cosmological constant Λ —a geometric property of the metric, uniform everywhere, which does not create structure; and (2) the displacement of matter by vacuum energy—a local field quantity, measurable in the laboratory, which exerts pressure on matter concentrations and drives the formation of cosmic voids and the cosmic web. These are different processes operating at different scales with different physics. Space expands uniformly (geometry). Vacuum pushes matter into clumps (field dynamics). In the early universe, when all energy—both vacuum and matter—was compressed into a smaller volume, the vacuum energy density was higher by a factor of $(1 + z)^3$. This denser vacuum exerted correspondingly stronger pressure on matter, assisting gravitational collapse and accelerating structure formation. As the universe expanded, the vacuum diluted, the pressure weakened, and the mechanism self-terminated—explaining why early structure formation was rapid and why it has slowed. The cosmological constant problem (the 120-order-of-magnitude discrepancy between the QFT vacuum energy and the observed Λ) is reframed in this approach because vacuum energy does not drive the expansion of space. Λ drives expansion. Vacuum energy drives structure. They are different quantities, they do different things, and they need not be equal. A further consequence emerges: inside gravitationally bound structures such as galaxies, expansion is suppressed, but vacuum gravity is not—producing an uncompensated gravitational contribution that matches the qualitative and quantitative description of dark matter.

Keywords: vacuum energy, cosmological constant, dark matter, early universe, structure formation, JWST, cosmic web, void formation, Casimir effect, rotation curves, gravitational lensing

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1 Introduction

The James Webb Space Telescope has revealed massive galaxies at redshifts that challenge the standard timeline. Labbé et al. (2023) found galaxies at $z > 10$ with stellar masses exceeding $10^{10} M_{\odot}$. Bogdán et al. (2024) detected a supermassive black hole at $z = 10.1$. Kokorev et al. (2024) identified compact red objects at $4 < z < 9$ hosting candidate AGN. Building these structures by $z = 10$ (~ 470 Myr) requires assembly rates 3–10 times faster than standard predictions (Boylan-Kolchin 2023).

This paper proposes that the puzzle arises from a conflation of two distinct physical processes:

1. **The expansion of space.** Driven by the cosmological constant Λ . Geometric. Uniform. Does not push matter around. Does not create structure.
2. **The displacement of matter by vacuum energy.** A local field effect. Vacuum energy is a form of matter—the ground state of quantum fields. It has energy, and by $E = mc^2$ it has mass, and by general relativity it gravitates.

Both principles are Einstein’s. The equivalence of energy and mass ($E = mc^2$, 1905) demands that vacuum energy gravitates. The cosmological constant (Λ , 1917) is a geometric property of spacetime—constant everywhere, constant always. We preserve both. We add nothing. We merely refuse to identify them.

The identification of Λ with vacuum energy was made by Zel’dovich (1967), not by Einstein. It was never proven. It produced a 10^{120} -order discrepancy between the QFT estimate and the observed value. We propose to undo it—returning Λ to what Einstein intended and letting vacuum energy be what it is: a form of matter that gravitates, displaces, and participates in structure formation.

2 Logical Structure of the Argument

Step 1: Λ and vacuum energy are different. Λ is global geometry, measured only cosmologically. Vacuum energy is local field physics, measured in laboratories. They have never agreed. (§3)

Step 2: Two processes, not one. Space expansion (Λ , geometry, uniform) and vacuum displacement of matter (local, creates structure). (§4)

Step 3: Vacuum energy was denser in the early universe. Same energy, smaller volume: $\rho_{\text{vac}}(z) \propto (1+z)^n$, $n > 0$. (§5)

Step 4: Denser vacuum pushed harder on matter. Stronger pressure on proto-galactic clouds from surrounding proto-voids. (§6)

Step 5: This accelerated structure formation. Gravity pulls, vacuum pushes. Two forces, one direction. (§7)

Step 6: The mechanism self-terminated. Vacuum diluted with expansion. (§8)

Step 7: Inside galaxies, vacuum gravity is uncompensated. Expansion is suppressed; vacuum gravitates; this is dark matter. (§9)

Step 8: The 120-order problem is reframed. Vacuum energy does not drive expansion. Λ does. They need not be equal. (§10)

3 Λ and Vacuum Energy Are Different

3.1 The Vacuum Has Structure and Energy

The quantum vacuum is not empty. The Standard Model is built on the premise that the vacuum has a nontrivial structure with nonzero energy content. The evidence comes from several independent lines:

(a) The Higgs field vacuum expectation value. The electroweak sector of the Standard Model requires the Higgs field to have a nonzero value in its ground state: $\langle\phi\rangle \approx 246\text{ GeV}$. This is not a perturbative correction; it is the foundation of the theory. The vacuum state of the Higgs field has a potential energy density set by the Mexican-hat potential. The measured Higgs boson mass ($m_H \approx 125\text{ GeV}$) confirms this structure experimentally.

(b) QCD vacuum condensates. The strong interaction generates a quark condensate $\langle\bar{q}q\rangle \neq 0$ and a gluon condensate $\langle G_{\mu\nu}G^{\mu\nu}\rangle \neq 0$ in the vacuum. These condensates are measured through QCD sum rules (Shifman, Vainshtein & Zakharov 1979) and lattice QCD calculations. The QCD vacuum energy density is of order $\sim -(0.2\text{ GeV})^4$. This is not speculative; it is required for the Standard Model to produce the observed hadron spectrum.

(c) The Casimir effect. The force between uncharged conducting plates (Casimir 1948), measured to sub-percent precision (Lamoreaux 1997; Bressi et al. 2002), demonstrates that modifying the boundary conditions of the vacuum produces a measurable force. We note that this effect can also be derived from van der Waals interactions without reference to zero-point energy (Jaffe 2005). The Casimir effect therefore establishes that the vacuum responds to geometry, but does not by itself prove that the absolute vacuum energy is nonzero.

(d) The Lamb shift and anomalous magnetic moment. The $2S_{1/2}$ – $2P_{1/2}$ splitting in hydrogen (Lamb & Retherford 1947) and the electron’s anomalous magnetic moment arise from higher-order virtual processes in QED. These demonstrate that the vacuum mediates interactions through its fluctuation structure, but—like the Casimir effect—they measure the consequences of vacuum fluctuations rather than the absolute energy of the ground state.

(e) Phase transitions. The electroweak phase transition changed the vacuum state of the universe at $T \sim 100\text{ GeV}$, and the QCD phase transition at $T \sim 150\text{ MeV}$. In each case, the vacuum energy density changed by a finite, calculable amount. If the vacuum energy were zero by construction, these transitions would have no energy cost—contradicting the Standard Model.

The strongest evidence for nonzero vacuum energy comes not from Casimir or Lamb but from (a), (b), and (e): the Higgs VEV, the QCD condensates, and the phase transitions. These are structural features of the Standard Model, not optional interpretations.

What remains unknown is the *absolute* gravitating density of the vacuum in its

present state. The QFT estimate ($\sim M_{\text{Pl}}^4$) is a naïve cutoff-dependent sum that almost no one regards as physical. The renormalized value is scheme-dependent. We do not know ρ_{vac} . But we know the vacuum is not energetically trivial.

3.2 Λ Is Global and Not Locally Measurable

Λ has never been measured in any laboratory. It is inferred only from cosmological observations: Type Ia supernovae (Riess et al. 1998; Perlmutter et al. 1999), BAO (DESI Collaboration 2024), and CMB (Planck Collaboration 2020). Every measurement of Λ is a measurement of global geometry. Einstein introduced it in 1917 without reference to quantum physics.

3.3 The 120-Order Discrepancy as Evidence of Non-Identity

If Λ and ρ_{vac} were the same, local measurements should give the global value. They do not: $\rho_{\text{vac}}^{\text{QFT}} \sim M_{\text{Pl}}^4 \sim 10^{76} \text{ GeV}^4$, while $\rho_{\Lambda}^{\text{obs}} \sim 10^{-47} \text{ GeV}^4$. A discrepancy of 10^{120} .

Standard interpretation: a theoretical crisis (Weinberg 1989).

Our interpretation: evidence that the two quantities are not the same. One is local. The other is global. They enter the Einstein equations at similar positions, which led to their identification by Zel’dovich (1967)—without proof. Weinberg (1989) framed the resulting discrepancy as the cosmological constant problem. Coleman (1988) and Kaloper & Padilla (2014) constructed mechanisms to decouple them. We argue the decoupling is the physically correct default.

Counterarguments and Responses

Objection: “In GR, any energy density gravitates. Vacuum energy must contribute to the Friedmann equation.”

Response: In QFT, absolute vacuum energy is unobservable; only differences are measurable. Standard renormalization subtracts the vacuum contribution. If this extends to gravity (as the sequestering programme proposes), vacuum energy does not enter the Friedmann equation. What enters is Λ —a geometric constant—and deviations from vacuum (matter, radiation). This is a proposal, not a theorem.

4 Two Processes, Not One

The standard narrative says: “Dark energy expands the universe and fills the voids.” We split this into two separate statements about two separate processes.

4.1 Process 1: Expansion of Space

Driven by Λ . Geometric. The metric itself stretches. This happens *everywhere uniformly*—inside galaxies, inside voids, inside atoms (though overwhelmed by local forces at small scales). It does not push matter around. It does not create structure. It does

not favor voids over walls. It is the property of the metric, not of anything *in* the metric.

4.2 Process 2: Vacuum Displaces Matter

Vacuum energy is a local field quantity: the energy of quantum field fluctuations in a given region. Where matter is present, the vacuum state is modified—Pauli blocking reduces fermionic modes, vacuum polarization alters the electromagnetic vacuum, curvature changes the mode spectrum (Kriger 2026). The vacuum energy density inside a matter concentration differs from that in the surrounding empty space.

The surrounding vacuum—unperturbed, at full energy density—exerts pressure on the matter concentration from all sides. This is the same physics as the Casimir effect: modified vacuum (between plates or inside matter) has different energy from unmodified vacuum (outside), and the energy difference produces a force. The force is directed inward: from high-vacuum-energy voids toward low-vacuum-energy matter concentrations.

This pressure pushes matter into clumps. The spaces left behind—emptied of matter—become voids. The voids are not “expanding” in the sense that Λ pushes them apart. They are *growing* because vacuum pressure continues to push residual matter toward the walls and filaments. The voids are filled with unperturbed vacuum, which is the energetically preferred state.

4.3 How Voids Grow: Two Contributions

A void grows for two reasons simultaneously:

1. **Space expansion.** Λ stretches the metric everywhere, including inside the void. This is geometric growth—the void gets bigger because space itself gets bigger.
2. **Vacuum displacement.** The vacuum pressure at the void-wall boundary pushes residual matter outward (from the void’s perspective) into the walls. This is physical growth—the void gets emptier because matter is expelled.

These are additive but independent processes. Process 1 happens whether or not there is any vacuum energy. Process 2 happens whether or not space is expanding. In the real universe, both occur simultaneously.

Counterarguments and Responses

Objection: “This distinction is semantic. In the Einstein equations, Λ and vacuum energy enter identically.”

Response: Mathematically, a constant $\Lambda g_{\mu\nu}$ on the left side and a constant $-\rho_{\text{vac}} g_{\mu\nu}$ on the right side produce the same equations. But ρ_{vac} is not constant—it depends on local conditions (Casimir effect). When the vacuum energy is inhomogeneous (because matter is inhomogeneous), its gradient produces forces that Λ cannot. The distinction is physical, not semantic: Λ has no gradient; $\rho_{\text{vac}}(\mathbf{x})$ does.

Section Result and Implications

Space expansion and vacuum displacement are different processes. Conflating them leads to confusion about what “dark energy does.” Separating them clarifies: Λ expands; vacuum pushes.

5 Vacuum Energy in the Early Universe

If vacuum energy is a local quantity—the energy of field fluctuations in a given volume—then in a smaller universe, the same energy occupied a smaller volume and the density was higher.

The exact scaling depends on the field content and regularization. For a comoving momentum cutoff, massless fields give $\rho_{\text{vac}} \propto a^{-4}$ (radiation-like), massive fields give $\propto a^{-3}$ (matter-like) (Parker & Toms 2009; Birrell & Davies 1982). We adopt $\rho_{\text{vac}}(z) \propto (1+z)^n$ with $n \approx 3$ as an illustrative assumption. The qualitative conclusion—vacuum was denser at high z —holds for any $n > 0$.

Epoch	Redshift z	$(1+z)^3$	$\rho_{\text{vac}}/\rho_{\text{vac},0}$
Today	0	1	1
Quasar peak	2	27	27
Cosmic web	5	216	216
JWST galaxies	12	2,197	$\sim 2,000$
First stars	20	9,261	$\sim 9,000$

Table 1: Vacuum energy density at selected epochs, assuming $n = 3$ scaling.

Does this vacuum energy enter the Friedmann equation? No—under our proposal. The Friedmann equation is driven by Λ (constant, geometric) and matter/radiation. The vacuum energy participates in *local* dynamics (pressure, displacement) but not in *global* expansion. This is the working hypothesis motivated by the sequestering programme (Kaloper & Padilla 2014). If this hypothesis is wrong, and vacuum energy with $w = 0$ does enter Friedmann, then Planck data require $\rho_{\text{vac},0} \ll \rho_{\Lambda}^{\text{obs}}$, weakening the mechanism. We identify the CMB test as the highest-priority constraint.

Counterarguments and Responses

Objection: “The vacuum is Lorentz-invariant with $w = -1$. It cannot dilute.”

Response: In flat Minkowski space, yes. In expanding FRW spacetime, Lorentz invariance is broken by the expansion; the mode spectrum evolves with $a(t)$ (Birrell & Davies 1982). The vacuum state at $a = 0.1$ differs from the state at $a = 1$. Assigning $w = 0$ is an effective description of this evolution.

Objection: “The $(1+z)^3$ scaling is assumed, not derived from the renormalized $\langle T_{\mu\nu} \rangle$.”

Response: Correct. A rigorous derivation from the renormalized stress-energy tensor of the Standard Model fields in FRW spacetime is an open problem. Our estimate

is parametric; the qualitative conclusion (denser vacuum at higher z) holds for any $n > 0$.

6 Vacuum Pressure on Early Structures

In the early universe, a proto-galactic cloud was surrounded by proto-voids containing unperturbed vacuum at full density. The vacuum pressure at the cloud boundary was:

$$\Delta P_{\text{vac}}(z) \sim \rho_{\text{vac}}(z) c^2 \sim \rho_{\text{vac},0} (1+z)^3 c^2. \quad (1)$$

At $z = 12$: $\Delta P_{\text{vac}} \sim 2,000 \rho_{\text{vac},0} c^2$.

Normalization caveat. The present vacuum energy density $\rho_{\text{vac},0}$ is unknown. If $\rho_{\text{vac},0} \sim \rho_{\Lambda}^{\text{obs}} \sim 6 \times 10^{-27} \text{ kg m}^{-3}$, the pressure is dynamically significant ($\sim 10^{-12} \text{ Pa}$, exceeding the gravitational self-pressure of proto-galactic clouds). If $\rho_{\text{vac},0}$ is many orders of magnitude smaller, the effect is negligible. All quantitative estimates are conditional on $\rho_{\text{vac},0}$.

The mechanism works because the vacuum pressure is *anisotropic at the cloud boundary*: full vacuum pressure on the void side, modified (reduced) vacuum on the matter side. The gradient drives matter inward—from voids into clumps. This is the same mechanism as the Casimir effect, where modified vacuum between plates has lower energy than unmodified vacuum outside, producing a net inward force.

Counterarguments and Responses

Objection: “Isotropic pressure cancels. You need to demonstrate gradient formation from the stress-energy tensor, not by analogy.”

Response: The pressure is isotropic within each region (void, cloud interior) but differs in magnitude between regions because the vacuum state differs. At the boundary, a pressure gradient exists—higher on the void side, lower inside. This is the same principle as atmospheric pressure acting on a vacuum chamber. The field-theoretic derivation of this gradient for realistic cosmological geometries is an open problem—the most important theoretical gap in this work.

7 Consequences for Early Structure Formation

At $z \sim 12$, a collapsing proto-galactic cloud experienced two inward-directed forces:

1. **Gravity.** Self-attraction of the matter cloud. Standard mechanism.
2. **Vacuum pressure.** External compression from the surrounding denser vacuum. $\sim 2,000\times$ stronger than today (conditional on normalization).

Both squeeze the cloud. The collapse was faster not because gravity was stronger, but because it had a partner.

Black hole seeds. Eddington-limited growth ($M \propto e^{t/t_{\text{Sal}}}$, $t_{\text{Sal}} \approx 45 \text{ Myr}$; Salpeter 1964) is exponentially sensitive to when the first seeds form. If vacuum pressure caused the first massive halos to collapse 30–50 Myr earlier, the exponential amplifies this

into orders of magnitude in final mass. This is a sensitivity illustration; quantitative predictions require simulations.

Giant early stars. Low metallicity prevents fragmentation (Bromm & Larson 2004). Vacuum pressure provides additional external compression, potentially producing more massive stars and heavier remnants.

8 Why the Mechanism Is Inoperative Today

1. Vacuum energy density at $z = 0$ is $\sim 2,000\times$ lower than at $z = 12$. Pressure proportionally weaker.
2. Present-day structures are far denser than the vacuum. Vacuum pressure is negligible compared to gravitational and gas forces.
3. The void-wall contrast in absolute terms is tiny because $\bar{\rho}_{m,0}$ is tiny.

This explains: no new giant galaxies forming today; declining quasar activity; no new supermassive black holes from scratch. The “second hand” has weakened below detectability.

Note: voids still grow today—but primarily through Process 1 (space expansion stretching them) rather than Process 2 (vacuum actively displacing matter). In the early universe, Process 2 was dominant.

9 Uncompensated Vacuum Gravity Inside Galaxies

The two-process framework leads to an unexpected consequence.

9.1 The Fundamental Observation

Every point in space contains energy. Energy is equivalent to mass ($E = mc^2$). Mass gravitates. This is general relativity, not a hypothesis.

The quantum vacuum has energy everywhere. Therefore it has mass everywhere. Therefore it gravitates everywhere. The question is not *whether* vacuum gravitates, but *whether its gravitation is observable*.

9.2 Two Phases of Vacuum

The answer depends on whether the vacuum is in the *expanding* phase or the *bound* phase:

Expanding phase (voids). In a void, the vacuum is uniform. There is no density gradient—no direction in which it can “fall.” Vacuum cannot collapse onto itself because it is the same everywhere; there is no center of attraction. Additionally, Λ stretches the space, compensating the vacuum’s gravitational self-attraction. The void is stable.

Bound phase (inside galaxies). Inside a gravitationally bound structure, the visible matter creates a potential well deep enough to prevent Λ from stretching the interior. Space inside a galaxy does not expand—this is standard physics (bound systems decouple from the Hubble flow). But the vacuum energy is still there, between

the stars, filling the vast empty spaces. And its gravity is no longer compensated by expansion, because there is no expansion here.

The vacuum inside the galaxy is *part of the galaxy*. It is gravitationally bound, just as the stars are. Its mass adds to the total. A star orbiting at the edge does not care whether the mass comes from stars or from vacuum. A photon being lensed does not care. Mass is mass.

9.3 The Phase Transition Boundary

The boundary between expanding and bound vacuum is the surface where the galaxy's gravitational potential can no longer overcome cosmic expansion. Inside: vacuum is bound, its gravity uncompensated, it is part of the galaxy. Outside: vacuum is free, expanding, its gravity compensated by Λ .

This boundary is the *dark matter halo edge*. Not an arbitrary cutoff but a physical phase transition. The boundary is smooth (the potential decreases gradually), producing a diffuse halo—exactly as observed.

And to infalling galaxy clusters, it does not matter what is inside the halo—stars, gas, or bound vacuum. They gravitate toward the total mass. Light bends around the total mass.

9.4 Order-of-Magnitude Estimate

A typical spiral galaxy has a visible (baryonic) mass of $M_{\text{bar}} \sim 5 \times 10^{10} M_{\odot}$ and a dark matter halo mass of $M_{\text{DM}} \sim 5 \times M_{\text{bar}} \sim 2.5 \times 10^{11} M_{\odot}$, enclosed within a radius $R \sim 100 \text{ kpc} \sim 3 \times 10^{21} \text{ m}$.

The volume of this region:

$$V = \frac{4}{3}\pi R^3 \approx 1.1 \times 10^{65} \text{ m}^3. \quad (2)$$

If the vacuum energy density is $\rho_{\text{vac},0}$ and it gravitates, the gravitating vacuum mass in this volume is:

$$M_{\text{vac}} = \rho_{\text{vac},0} V. \quad (3)$$

For M_{vac} to match M_{DM} :

$$\rho_{\text{vac},0} = \frac{M_{\text{DM}}}{V} = \frac{2.5 \times 10^{11} \times 2 \times 10^{30}}{1.1 \times 10^{65}} \approx 4.5 \times 10^{-24} \text{ kg m}^{-3}. \quad (4)$$

Compare with the observed dark energy density: $\rho_{\Lambda}^{\text{obs}} \approx 5.9 \times 10^{-27} \text{ kg m}^{-3}$. The required $\rho_{\text{vac},0}$ is ~ 800 times larger than $\rho_{\Lambda}^{\text{obs}}$.

This is a discrepancy—but a discrepancy of a factor of 10^3 , not 10^{120} . And the comparison is not with Λ (which is geometry, not vacuum energy) but with whatever the local vacuum energy density actually is. If $\rho_{\text{vac},0}$ is set by the physics of vacuum fluctuations rather than by Λ , there is no a priori reason it should equal $\rho_{\Lambda}^{\text{obs}}$.

9.5 Rotation Curve Profile

The gravitating vacuum is uniformly distributed (it fills the volume between stars). A uniform mass density ρ_{vac} inside a sphere of radius r contributes:

$$M_{\text{vac}}(r) = \frac{4}{3}\pi \rho_{\text{vac}} r^3. \quad (5)$$

The circular velocity contribution from this uniform component:

$$v_{\text{vac}}(r) = \sqrt{\frac{G M_{\text{vac}}(r)}{r}} = \sqrt{\frac{4\pi G \rho_{\text{vac}}}{3}} r. \quad (6)$$

This rises linearly with r . Combined with the falling Keplerian contribution from the baryonic disk ($v_{\text{bar}} \propto r^{-1/2}$ at large r), the total rotation curve:

$$v_{\text{tot}}(r) = \sqrt{v_{\text{bar}}^2(r) + v_{\text{vac}}^2(r)} \quad (7)$$

transitions from rising (inner region, baryonic disk dominates) to approximately flat (outer region, rising vacuum contribution compensates falling baryonic contribution)—exactly as observed.

This is qualitatively the same as an NFW profile at intermediate radii. A detailed comparison with observed rotation curves for specific galaxies is beyond the scope of this paper but constitutes a concrete, falsifiable prediction.

9.6 Gravitational Lensing

A photon passing through a galaxy is deflected by the total gravitational mass, including the vacuum contribution. Since the vacuum mass is distributed more uniformly than the baryonic mass (which is concentrated in a disk and bulge), the lensing profile would differ from a point-mass prediction—stronger at large impact parameters than the visible mass alone would produce. This is consistent with the observed lensing signatures attributed to dark matter halos.

9.7 The Bullet Cluster

The Bullet Cluster (1E 0657-56) is often cited as the strongest evidence for particulate dark matter. Two galaxy clusters collided; the hot gas (detected in X-rays) was slowed by ram pressure and remains near the collision center, while the gravitational mass (detected by lensing) has passed through and is offset from the gas.

This observation is qualitatively consistent with gravitating vacuum. Gas is made of particles—it collides. Vacuum is a field—it cannot collide with itself. Two regions of vacuum simply overlap; there is no scattering cross-section, no ram pressure, no friction. When two clusters merge, their bound vacuum passes through the opposing bound vacuum without interaction, as the lensing data require.

However, a quantitative test is more demanding. If vacuum energy density depends on local curvature (as Section 9.5 requires for rotation curves), then during a cluster

merger the vacuum energy would need to track the evolving gravitational potential in a complex, geometry-dependent way. Whether this reproduces the observed lensing morphology—two distinct mass concentrations spatially offset from the gas—is an open question that requires detailed modelling. We note this as a necessary future test, not a confirmed prediction.

Counterarguments and Responses

Objection: “Dark matter must be cold and collisionless to reproduce the CMB power spectrum and the large-scale structure.”

Response: The vacuum is “cold” in the sense that it has no bulk velocity (it fills space uniformly). It is “collisionless” in the sense that it does not scatter—it is a field, not particles. Both properties that CMB fits require of dark matter are automatically satisfied by gravitating vacuum. Whether the vacuum can reproduce the specific CMB acoustic peak ratios (which constrain $\Omega_{\text{DM}}h^2$ to $\sim 1\%$) requires a detailed calculation that has not yet been performed.

Objection: “The required $\rho_{\text{vac},0}$ is $\sim 800\times$ the observed dark energy density. This is another fine-tuning.”

Response: It is a discrepancy of $\sim 10^3$, not 10^{120} . And the comparison is illegitimate if $\Lambda \neq \rho_{\text{vac}}$. The vacuum energy density is whatever it is; it need not equal $\Lambda/8\pi G$. Whether $\rho_{\text{vac},0} \sim 5 \times 10^{-24} \text{ kg m}^{-3}$ is a “natural” value depends on the underlying physics, which is currently unknown.

Objection: “If vacuum gravity explains dark matter, you have eliminated one mystery but created another: what sets $\rho_{\text{vac},0}$?”

Response: Correct. But the same objection applies to dark matter particles: what sets their mass and cross-section? Decades of direct detection experiments have failed to find them. We replace an undetected particle with a measured phenomenon (vacuum energy)—at the cost of not knowing its gravitating density.

Objection: “Vacuum energy is the same everywhere. How can it form a halo?”

Response: The vacuum energy density *is* the same everywhere. What differs is whether its gravity is compensated by expansion. Inside the bound phase: uncompensated, contributes to the gravitational mass. Outside: compensated by Λ , gravitationally invisible. The “halo” is not a concentration of vacuum—it is the region where vacuum gravity is uncompensated. Its edge is the phase transition between bound and expanding vacuum.

Section Result and Implications

If vacuum energy gravitates ($E = mc^2$, GR) and expansion is absent inside bound structures, the vacuum acts as a uniformly distributed, invisible, collisionless gravitating component—matching the description of dark matter. The halo boundary is a phase transition between bound and expanding vacuum. The required density ($\sim 5 \times 10^{-24} \text{ kg m}^{-3}$) is empirically testable. If confirmed, dark matter is not a particle—it is vacuum energy, gravitationally bound to the structures it helps hold together.

10 The 120-Order Problem Is Reframed

In this framework, the cosmological constant problem is reframed:

- Λ is a geometric property of spacetime. It drives expansion. Its value is $\rho_{\Lambda}^{\text{obs}} \sim 10^{-47} \text{ GeV}^4$. This is a measured number.
- ρ_{vac} is the energy of quantum field fluctuations. It drives structure formation (via pressure and displacement). Its value is whatever QFT computes—possibly $\sim M_{\text{Pl}}^4 \sim 10^{76} \text{ GeV}^4$.
- They do different things. They need not be equal. There is no problem.

The “problem” arose because in 1967 Zel’dovich identified ρ_{vac} with Λ . This identification was never proven. It was motivated by the mathematical observation that both enter the Einstein equations as terms proportional to $g_{\mu\nu}$. But this mathematical similarity does not imply physical identity, any more than two forces that both obey $F \propto 1/r^2$ are the same force.

The vacuum energy may be enormous. It may be 10^{120} times Λ . This is not a problem—it is simply two different quantities having different values. One of them (vacuum energy) pushes matter around locally and gravitates inside bound structures. The other (Λ) stretches space globally. Neither needs to explain the other.

In a sense, Einstein was right—twice. His $E = mc^2$ (1905) demands that vacuum energy gravitates. His Λ (1917) is a geometric constant of spacetime. Both stand. What does not stand is the identification $\Lambda = 8\pi G\rho_{\text{vac}}$, introduced by Zel’dovich (1967), which was never a consequence of Einstein’s theory but an additional assumption. The 120-order “crisis” is the result of this assumption. Remove it, and both of Einstein’s results are preserved exactly as stated—and three major puzzles (early galaxies, dark matter, the cosmological constant problem) find a common resolution.

Counterarguments and Responses

Objection: “This doesn’t *solve* the CC problem. It just refuses to acknowledge it.”

Response: The CC problem as traditionally stated—why do Λ and ρ_{vac} disagree by 10^{120} ?—exists only if $\Lambda = 8\pi G\rho_{\text{vac}}$. If this identity is wrong, that specific question loses its force. However, the residual question—why doesn’t the large vacuum energy dominate cosmological expansion?—remains. We do not solve this; we reframe it as a separate problem that may have a separate answer (e.g., vacuum sequestering; Kaloper & Padilla 2014). The reframing may or may not be productive, but the original identification was assumed without proof, and questioning it is legitimate.

11 Connections to Existing Work

11.1 Vacuum Sequestering

Weinberg (1989) surveyed mechanisms for neutralizing vacuum gravity. Coleman (1988) proposed wormhole-mediated cancellation. Kaloper & Padilla (2014) built Lagrangian-level sequestering. Our proposal is consistent with this tradition: vacuum energy does not enter the Friedmann equation.

11.2 Backreaction Programme

Buchert (2000) and Räsänen (2004) showed that averaging over inhomogeneities produces effective dark-energy-like terms. Wiltshire (2007) attributed apparent acceleration to differential expansion. Seifert et al. (2025) found evidence for this in supernova data. Our model differs: we do not derive effective dark energy from averaging. We propose that vacuum energy is a separate dynamical component that drives structure formation locally. The two approaches may be complementary.

11.3 Penrose and Ellis

Penrose’s gravitational entropy programme (1979, 2004, 2010): the growth of structure corresponds to increasing Weyl curvature. Our mechanism provides a physical driver—vacuum pressure—for rapid early structure growth. Ellis’s fitting problem (1984, 1987): if vacuum energy is local and inhomogeneous, fitting to a smooth Friedmann model introduces systematic errors. Both connections are qualitative.

11.4 Companion Paper

A companion paper (Kriger 2026) developed a parametric model of spatial vacuum energy variation: $\Lambda(\rho_m) = \Lambda_0 - \alpha\rho_m$, addressing the void-vs-wall contrast at a fixed epoch. The present paper addresses temporal evolution. The two approaches are complementary but independent.

12 Discussion

12.1 The Central Claim

If Λ and ρ_{vac} are different, three things follow immediately:

1. Vacuum energy was denser in the past (it dilutes; Λ does not).
2. Vacuum pressure on matter was stronger in the early universe.
3. The 120-order problem is reframed (it ceases to be a comparison of two supposedly identical quantities).

If $\Lambda = \rho_{\text{vac}}$, the model fails entirely.

12.2 The Normalization Problem

$\rho_{\text{vac},0}$ is unknown. If comparable to $\rho_{\Lambda}^{\text{obs}}$, the vacuum pressure was dynamically significant. If much smaller, the effect is negligible. This is the model’s largest uncertainty—spanning potentially many orders of magnitude. Determining $\rho_{\text{vac},0}$ independently (perhaps through precision Casimir measurements of absolute vacuum energy, or through lattice QFT calculations) would test the model.

12.3 The CMB Test

Compatibility with the CMB is not an optional follow-up; it is a prerequisite for viability. The baryon-to-dark-matter ratio at recombination is measured to high precision by Planck ($\Omega_{\text{DM}}h^2 = 0.120 \pm 0.001$). Dark matter must behave as a pressureless, clustering component before decoupling. If vacuum energy with $w \approx 0$ enters the Friedmann equation, Planck data require $\rho_{\text{vac},0} \ll \rho_{\Lambda}^{\text{obs}}$, severely weakening the mechanism. Under the sequestering hypothesis, vacuum energy is gravitationally inert at the global level and no CMB conflict arises—but this hypothesis must be tested. Albareti & Maroto (2014) showed that vacuum energy of massive fields can scale as nonrelativistic matter and that vacuum density perturbations grow on sub-Hubble scales; whether this extends to the full Standard Model field content is unknown. A rigorous MCMC analysis is the highest priority for future work and could be decisive.

13 Limitations

1. The separation of Λ and ρ_{vac} is a proposal, not a proven fact.
2. The scaling $\rho_{\text{vac}} \propto (1+z)^n$ is parametric; n is model-dependent.
3. $\rho_{\text{vac},0}$ is unknown. Quantitative predictions are conditional.
4. The pressure gradient mechanism is Casimir-motivated, not derived from the stress-energy tensor in cosmological geometry.
5. If vacuum energy enters Friedmann, Planck constraints may be severe.
6. No simulations performed.
7. Hawking radiation is theoretical, not directly observed.
8. Connections to Penrose/Ellis/backreaction are qualitative.

14 Future Directions

1. Derivation of $\langle T_{\mu\nu} \rangle_{\text{ren}}$ in FRW for Standard Model fields.
2. Hydrodynamic simulations with vacuum pressure.
3. MCMC fit to Planck with diluting vacuum component.
4. Independent determination of $\rho_{\text{vac},0}$.
5. Modified Press–Schechter with vacuum pressure.
6. Lagrangian formulation of the two-process separation.

15 Conclusions

We have proposed that three major puzzles in cosmology—early galaxy formation, dark matter, and the cosmological constant problem—share a common origin: the conflation of two distinct quantities (Λ and ρ_{vac}) and two distinct processes (space expansion and vacuum displacement of matter).

Λ is global geometry. It stretches space uniformly. It is constant everywhere and always—exactly as Einstein intended in 1917.

Vacuum energy is local field physics. It has mass ($E = mc^2$). It gravitates. It pushes matter into clumps, forming the cosmic web. In the early universe, it was $\sim 2,000$ times denser and pushed $\sim 2,000$ times harder, accelerating structure formation. As the universe expanded, it diluted and the mechanism turned off.

Inside galaxies, where expansion is suppressed by gravitational binding, vacuum gravity is uncompensated—producing the invisible, collisionless, uniformly distributed gravitating component we call dark matter. The dark matter halo boundary is a phase transition between bound and expanding vacuum. Dark matter “passes through itself” (Bullet Cluster) because fields superpose—they do not collide.

The 120-order problem is reframed: if vacuum energy and Λ are different quantities with different roles, they need not be equal.

The model is falsifiable: if $\Lambda = \rho_{\text{vac}}$ can be demonstrated, it fails. If simulations cannot reproduce observed rotation curves with uniform vacuum density, it fails. If the required $\rho_{\text{vac},0} \sim 5 \times 10^{-24} \text{ kg m}^{-3}$ is excluded by independent measurement, it fails.

16 Peer Reviews and Revision History

First Round

Reviewer assessment (multiple reviewers). The paper was found to be bold, clearly written, and to address timely issues. Major concerns raised: (1) the $(1+z)^3$ scaling is assumed, not derived from the renormalized $\langle T_{\mu\nu} \rangle$; (2) the normalization $\rho_{\text{vac},0}$ is unknown, making all quantitative estimates conditional; (3) CMB compatibility is a prerequisite, not an optional test; (4) the vacuum pressure gradient is motivated by Casimir analogy but not derived from the stress-energy tensor in cosmological geometry; (5) claiming the CC problem “dissolves” overclaims; (6) the Bullet Cluster argument was too casual; (7) self-citation balance. Verdict: major revisions.

Response. (1) The scaling is now explicitly labeled as a parametric assumption with model-dependent exponent n ; qualitative conclusions hold for any $n > 0$. (2) The normalization uncertainty is emphasized throughout as the single largest uncertainty. (3) The CMB section has been rewritten to state that compatibility is a prerequisite, not a follow-up, and cites Albareti & Maroto (2014) on vacuum energy behaving as nonrelativistic matter. (4) The pressure gradient is acknowledged as the most important theoretical gap; we do not claim a derivation. (5) “Dissolves” has been replaced by “is reframed,” with explicit discussion of the residual question. (6) The Bullet Cluster section now acknowledges that a quantitative lensing model is needed. (7) The companion paper citation has been toned down.

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A Computational Code

```
import numpy as np

rho_vac_0 = 5.9e-27 # kg/m^3 (illustrative: ~ rho_Lambda_obs)
c = 3e8 # m/s
G = 6.674e-11 # m^3 kg^-1 s^-2

def rho_vac(z, n=3):
    return rho_vac_0 * (1+z)**n

def P_vac(z, n=3):
    return rho_vac(z, n) * c**2

# Table
print(f"{'z':>5} {'rho/rho0':>12} {'P_vac (Pa)':>12}")
for z in [0, 2, 5, 12, 20]:
    print(f"{z:5d} {(1+z)**3:12.0f} {P_vac(z):12.2e}")

# Comparison at z=12
rho_cloud = 100 * 2000 * 2.7e-27 # 100x overdensity at z=12
R = 3e19 # 1 kpc
P_grav = G * rho_cloud**2 * R**2
P_v = P_vac(12)
print(f"\nz=12: P_vac={P_v:.2e} Pa, P_grav={P_grav:.2e} Pa")
print(f"Ratio: {P_v/P_grav:.0f}")
print("(Conditional on rho_vac_0 ~ rho_Lambda_obs)")
```